**DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS**

# Mathematics II  Advanced Calculus (2AC)

**Exercise Sheet (for your practice):** **March 2023**

1. Say whether the limit exists and possibly find its value:

   (a) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{xy}{|x|+|y|}$;  (b) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{\sin(x^4y^2)}{x^4+y^2}$;  (c) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{e^{(x^2+y^4)}-1}{x^4y^4}$;

   (d) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{e^{(x^2+y^4)}-1}{|xy|}$;  (e) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{xy\sqrt{|xy|}}{x^2+xy+y^2}$.

   Hints: $\displaystyle\lim_{t\to 0}\frac{\sin t}{t} = ??$; $\displaystyle\lim_{t\to 0}\frac{e^t-1}{t} = ??$; $x^2+xy+y^2 \geq 0$; compare $|xy|$ and $x^2+xy+y^2$.

2. Let $\begin{cases} A := \{(x,y)\in\mathbb{R}^2\colon\ x^2 \leq y-1\}, \quad B := \{(x,y)\in\mathbb{R}^2\colon\ y \leq 1\}, \\ C := \{(x,y)\in\mathbb{R}^2\colon\ 1 < y < x^2+1\}. \end{cases}$.

   Let $f:\mathbb{R}^2\to\mathbb{R}$ be defined by $f(x,y) := \begin{cases} 0 & \text{if } (x,y)\in A\cup B, \\ 1 & \text{if } (x,y)\in C, \end{cases}$ for every $(x,y)\in\mathbb{R}^2$.

   (i) Study the continuity of the function $f$ at the point $(0,1)$.
   (ii) Find $f_x(0,1)$ and $f_y(0,1)$.
   (iii) Study the differentiability of the function $f$ at the point $(0,1)$.

3. Let $\begin{cases} E := \{(x,y)\in\mathbb{R}^2\colon\ \alpha x \leq y \leq \beta x\} \text{ with } 0<\alpha<\beta; \\ F := \{(x,y)\in\mathbb{R}^2\colon\ x>0,\ y>0\}. \end{cases}$

   (a) Explain why the domain $E$ lies in the first quadrant.
   (b) Consider the function $f(x,y) := \dfrac{x^5+y^5}{x^2y^2}$.
   Say whether the limits below exist and possibly find them:

   (i) $\displaystyle\lim_{\substack{(x,y)\to(0,0)\\(x,y)\in E}} f(x,y)$;  (ii) $\displaystyle\lim_{\substack{(x,y)\to(0,0)\\(x,y)\in F}} f(x,y)$.

   Hint: (ii) You can consider the parabolas $y=\alpha x^2$ with $0<x<1$ or $x=\beta y^2$ with $0<y<1$.

4. Let $G_1 := \{(x,y)\in\mathbb{R}^2\colon\ y\geq\lambda|x|,\ \lambda>1\}$ and $G_2 := \{(x,y)\in\mathbb{R}^2\colon\ y>|x|\}$.

   Consider the function $g(x,y) := \dfrac{x^2y}{y^2-x^2}$. Find the limits:

   (a) $\displaystyle\lim_{\substack{(x,y)\to(0,0)\\(x,y)\in G_1}} g(x,y)$;  (b) $\displaystyle\lim_{\substack{(x,y)\to(0,0)\\(x,y)\in G_2}} g(x,y)$.

Hint: (b) You can consider for instance the arc of curve $y = |x| + x^2$ for $|x| > 0$ and definitely small. Then, use (a) to conclude.

5. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x,y) := \begin{cases} \dfrac{x^3 - y^3}{2x^2 + xy + y^2} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$

   (i) Study the continuity of the function $f$ at the point $(0,0)$.
   Hint: you can use for instance $1 + \cos^2(\theta) + \sin\theta\cos\theta = 1 + \cos^2(\theta) + \frac{1}{2}\sin(2\theta) \geq ????$.

   (ii) Find $f_x(0,0)$ and $f_y(0,0)$.

   (iii) Study the differentiability of the function $f$ at the point $(0,0)$.
   Hint: Use the definition of differentiability and check the limit along the line $y = x$.